

DATA STRUCTURE

LAB PROGRAMS

6. a) Write a C program for insertion of a newnode at the beginning of a single linked list .

```
#include <stdio.h>

#include <stdlib.h>

struct Node

{

    int data;

    struct Node *next;

};

void display(struct Node *head)

{

    struct Node *temp = head;

    while(temp != NULL)

    {

        printf("%d -> ", temp->data);

        temp = temp->next;

    }

    printf("NULL\n");

}

int main()

{

    struct Node *head, *newNode;

    head = (struct Node *)malloc(sizeof(struct Node));

    head->data = 10;
```

```
head->next = (struct Node *)malloc(sizeof(struct Node));
head->next->data = 20;
head->next->next = (struct Node *)malloc(sizeof(struct Node));
head->next->next->data = 30;
head->next->next->next = NULL;
printf("Before Insertion:\n");
display(head);
newNode = (struct Node *)malloc(sizeof(struct Node));
newNode->data = 5;
newNode->next = head;
head = newNode;
printf("\nAfter Insertion at Beginning:\n");
display(head);
return 0;
}
```

Output

Before Insertion:

10 -> 20 -> 30 -> NULL

After Insertion at Beginning:

5 -> 10 -> 20 -> 30 -> NULL

=== Code Execution Successful ===

b) Write a C program for insertion of a newnode at the end of a single linked list .

```
#include <stdio.h>

#include <stdlib.h>

struct Node

{

    int data;

    struct Node *next;

};

void display(struct Node *head)

{

    struct Node *temp = head;

    while(temp != NULL)

    {

        printf("%d -> ", temp->data);

        temp = temp->next;

    }

    printf("NULL\n");

}

int main()

{

    struct Node *head, *temp, *newNode;

    head = (struct Node *)malloc(sizeof(struct Node));

    head->data = 10;

    head->next = (struct Node *)malloc(sizeof(struct Node));

    head->next->data = 20;

    head->next->next = (struct Node *)malloc(sizeof(struct Node));

    head->next->next->data = 30;
```

```
head->next->next->next = NULL;
printf("Before Insertion:\n");
display(head);
newNode = (struct Node *)malloc(sizeof(struct Node));
newNode->data = 40;
newNode->next = NULL;
temp = head;
while(temp->next != NULL)
{
    temp = temp->next;
}
temp->next = newNode;
printf("\nAfter Insertion at End:\n");
display(head);
return 0;
}
```

Output

Before Insertion:

10 -> 20 -> 30 -> NULL

After Insertion at End:

10 -> 20 -> 30 -> 40 -> NULL

=== Code Execution Successful ===

c) Write a C program for insertion of a newnode at any position of a single linked list .

```
#include <stdio.h>

#include <stdlib.h>

struct Node

{

    int data;

    struct Node *next;

};

void display(struct Node *head)

{

    struct Node *temp = head;

    while(temp != NULL)

    {

        printf("%d -> ", temp->data);

        temp = temp->next;

    }

    printf("NULL\n");

}

int main()

{

    struct Node *head, *temp, *newNode;

    int pos, i;

    head = (struct Node *)malloc(sizeof(struct Node));

    head->data = 10;

    head->next = (struct Node *)malloc(sizeof(struct Node));

    head->next->data = 20;

    head->next->next = (struct Node *)malloc(sizeof(struct Node));

    head->next->next->data = 30;
```

```
head->next->next->next = NULL;
printf("Before Insertion:\n");
display(head);
newNode = (struct Node *)malloc(sizeof(struct Node));
newNode->data = 25;
pos = 3;
temp = head;
for(i = 1; i < pos - 1; i++)
{
    temp = temp->next;
}
newNode->next = temp->next;
temp->next = newNode;
printf("\nAfter Insertion at Position %d:\n", pos);
display(head);
return 0;
}
```

Output

Before Insertion:

10 -> 20 -> 30 -> NULL

After Insertion at Position 3:

10 -> 20 -> 25 -> 30 -> NULL

=== Code Execution Successful ===